

Lecture Notes: Sequences and Limits

Calculus I

Author: Bakdaulet Sotsial

Astana IT University

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Section 1 Sequences and Their Limits

Definition 1 (Sequence) A sequence is an ordered list of numbers, denoted by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$, where each number a_n is called a term. The subscript n is the index, usually a positive integer. A sequence can be thought of as a function whose domain is the set of positive integers.

Example 1 Some examples of sequences:

$$1. a_n = \frac{1}{n}: 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

$$2. a_n = (-1)^n: -1, 1, -1, 1, \dots$$

$$3. a_n = \frac{n}{n+1}: \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$$

$$4. a_n = 2^n: 2, 4, 8, 16, \dots$$

Definition 2 (Limit of a Sequence) A sequence $\{a_n\}$ converges to a limit L , written as

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \text{ as } n \rightarrow \infty,$$

if for every $\epsilon > 0$ there exists a positive integer N such that for all $n > N$, we have $|a_n - L| < \epsilon$. If no such L exists, the sequence diverges.

Example 2 Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Let $\epsilon > 0$ be given. We need to find N such that for all $n > N$, $|\frac{1}{n} - 0| < \epsilon$. This inequality is equivalent to $\frac{1}{n} < \epsilon$, or $n > \frac{1}{\epsilon}$. Choose $N = \lceil \frac{1}{\epsilon} \rceil$. Then for all $n > N$, we have $n > \frac{1}{\epsilon}$, so $\frac{1}{n} < \epsilon$. Thus, $|\frac{1}{n} - 0| < \epsilon$. Hence the limit is 0.

Definition 3 (Divergence to Infinity) A sequence $\{a_n\}$ diverges to infinity, written $\lim_{n \rightarrow \infty} a_n = \infty$, if for every $M > 0$ there exists N such that for all $n > N$, $a_n > M$. Similarly, $\lim_{n \rightarrow \infty} a_n = -\infty$ if for every $M > 0$ there exists N such that for all $n > N$, $a_n < -M$.

Example 3 The sequence $a_n = n$ diverges to infinity. The sequence $a_n = (-1)^n$ diverges but not to infinity; it oscillates.

Section 2 Properties of Convergent Sequences

Theorem 1 (Uniqueness of Limit) If a sequence converges, its limit is unique.

Suppose $\{a_n\}$ converges to both L_1 and L_2 with $L_1 \neq L_2$. Let $\epsilon = \frac{|L_1 - L_2|}{2} > 0$. Since $a_n \rightarrow L_1$, there exists N_1 such that for $n > N_1$, $|a_n - L_1| < \epsilon$. Since $a_n \rightarrow L_2$, there exists N_2 such that for $n > N_2$, $|a_n - L_2| < \epsilon$. Let $N = \max(N_1, N_2)$. For $n > N$, both inequalities hold. Then

$$|L_1 - L_2| = |L_1 - a_n + a_n - L_2| \leq |L_1 - a_n| + |a_n - L_2| < \epsilon + \epsilon = 2\epsilon = |L_1 - L_2|,$$

a contradiction. Hence $L_1 = L_2$.

Theorem 2 (Algebraic Limit Theorem) If $\{a_n\}$ and $\{b_n\}$ are convergent sequences with $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$, then:

1. $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$.
2. $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$.
3. $\lim_{n \rightarrow \infty} (ca_n) = cA$ for any constant c .

$$4. \lim_{n \rightarrow \infty} (a_n b_n) = AB.$$

$$5. \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}, \text{ provided } B \neq 0 \text{ and } b_n \neq 0 \text{ for all } n.$$

We prove (1) and (4); the others are similar. For (1): Let $\epsilon > 0$. Choose N_1 such that $|a_n - A| < \epsilon/2$ for $n > N_1$, and N_2 such that $|b_n - B| < \epsilon/2$ for $n > N_2$. Let $N = \max(N_1, N_2)$. Then for $n > N$,

$$|(a_n + b_n) - (A + B)| \leq |a_n - A| + |b_n - B| < \epsilon/2 + \epsilon/2 = \epsilon.$$

For (4): Note that

$$|a_n b_n - AB| = |a_n b_n - a_n B + a_n B - AB| \leq |a_n| |b_n - B| + |B| |a_n - A|.$$

Since $\{a_n\}$ converges, it is bounded, so there exists $M > 0$ such that $|a_n| \leq M$ for all n . Let $\epsilon > 0$. Choose N_1 such that $|a_n - A| < \frac{\epsilon}{2(|B|+1)}$ for $n > N_1$, and N_2 such that $|b_n - B| < \frac{\epsilon}{2M}$ for $n > N_2$. Let $N = \max(N_1, N_2)$. Then for $n > N$,

$$|a_n b_n - AB| \leq M \cdot \frac{\epsilon}{2M} + |B| \cdot \frac{\epsilon}{2(|B|+1)} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Theorem 3 (Squeeze Theorem for Sequences) If $a_n \leq b_n \leq c_n$ for all n beyond some index, and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Example 4 Find $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$. Since $-1 \leq \sin n \leq 1$ for all n , we have $-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$. Both $-\frac{1}{n}$ and $\frac{1}{n}$ converge to 0. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} \frac{\sin n}{n} = 0$.

Section 3 Monotone Sequences

Definition 4 (Monotone Sequence) A sequence $\{a_n\}$ is called:

- increasing if $a_n \leq a_{n+1}$ for all n .
- decreasing if $a_n \geq a_{n+1}$ for all n .
- strictly increasing if $a_n < a_{n+1}$ for all n .
- strictly decreasing if $a_n > a_{n+1}$ for all n .

A sequence that is either increasing or decreasing is called monotone.

Definition 5 (Bounded Sequence) A sequence $\{a_n\}$ is bounded above if there exists a number M such that $a_n \leq M$ for all n . It is bounded below if there exists a number m such that $a_n \geq m$ for all n . It is bounded if it is both bounded above and below.

Theorem 4 (Monotone Convergence Theorem) Every bounded, monotone sequence is convergent. Specifically:

- If $\{a_n\}$ is increasing and bounded above, then it converges to its least upper bound.
- If $\{a_n\}$ is decreasing and bounded below, then it converges to its greatest lower bound.

We prove the increasing case. Let $S = \{a_n : n \in \mathbb{N}\}$ and let $L = \sup S$. Since the sequence is bounded above, L exists. We claim $a_n \rightarrow L$. Let $\epsilon > 0$. Since L is the least upper bound, $L - \epsilon$ is not an upper bound, so there exists N such that $a_N > L - \epsilon$. Since the sequence is increasing, for all $n > N$, we have $a_n \geq a_N > L - \epsilon$. Also, since L is an upper bound, $a_n \leq L$ for all n . Thus for $n > N$, $L - \epsilon < a_n \leq L < L + \epsilon$, so $|a_n - L| < \epsilon$. Hence $a_n \rightarrow L$.

Example 5 The sequence $a_n = 1 - \frac{1}{n}$ is increasing and bounded above by 1. By the Monotone Convergence Theorem, it converges. Its limit is 1.

Example 6 The sequence $a_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded above by 3. It converges to Euler's number $e \approx 2.71828$.

Section 4 Limits of Functions

Definition 6 (Limit of a Function) Let f be a function defined on an open interval containing a , except possibly at a itself. We say that the limit of $f(x)$ as x approaches a is L , written

$$\lim_{x \rightarrow a} f(x) = L,$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.

Example 7 Show that $\lim_{x \rightarrow 2} (3x + 1) = 7$. Let $\epsilon > 0$. We need δ such that if $0 < |x - 2| < \delta$, then $|(3x + 1) - 7| < \epsilon$. Simplify: $|3x - 6| = 3|x - 2| < \epsilon$, so $|x - 2| < \epsilon/3$. Choose $\delta = \epsilon/3$. Then if $0 < |x - 2| < \delta$, we have $|3x + 1 - 7| = 3|x - 2| < 3\delta = \epsilon$. Hence the limit is 7.

Definition 7 (One-Sided Limits) The left-hand limit of $f(x)$ as x approaches a from the left is L , written $\lim_{x \rightarrow a^-} f(x) = L$, if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $a - \delta < x < a$, then $|f(x) - L| < \epsilon$. The right-hand limit $\lim_{x \rightarrow a^+} f(x) = L$ is defined similarly with $a < x < a + \delta$.

Theorem 5 $\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$.

Example 8 Consider $f(x) = \begin{cases} x + 1 & \text{if } x < 0 \\ x^2 + 2 & \text{if } x \geq 0 \end{cases}$. Find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$. As $x \rightarrow 0^-$, $f(x) = x + 1 \rightarrow 1$. As $x \rightarrow 0^+$, $f(x) = x^2 + 2 \rightarrow 2$. Since the one-sided limits are different, the two-sided limit does not exist.

Theorem 6 (Algebraic Limit Laws for Functions) If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$.
2. $\lim_{x \rightarrow a} (f(x) - g(x)) = L - M$.
3. $\lim_{x \rightarrow a} (cf(x)) = cL$.
4. $\lim_{x \rightarrow a} (f(x)g(x)) = LM$.
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$.

Section 5 The Precise Definition of a Limit

The precise definition, also known as the epsilon-delta definition, is the foundation of calculus. We restate it clearly:

Definition 8 (Epsilon-Delta Definition) Let f be a function defined on an open interval containing a , except possibly at a . We say that $\lim_{x \rightarrow a} f(x) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

Example 9 Prove that $\lim_{x \rightarrow 3} x^2 = 9$. Let $\epsilon > 0$. We need to find $\delta > 0$ such that if $0 < |x - 3| < \delta$, then $|x^2 - 9| < \epsilon$. Note that $|x^2 - 9| = |x - 3||x + 3|$. We can bound $|x + 3|$ by restricting x to be close to 3. Suppose $|x - 3| < 1$, then $2 < x < 4$, so $5 < x + 3 < 7$, hence $|x + 3| < 7$. Then $|x^2 - 9| < 7|x - 3|$. To make this less than ϵ , choose $|x - 3| < \epsilon/7$. Let $\delta = \min(1, \epsilon/7)$. Then if $0 < |x - 3| < \delta$, we have $|x^2 - 9| < 7 \cdot \frac{\epsilon}{7} = \epsilon$. Thus the limit is 9.

Theorem 7 (Uniqueness of Limits) If $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} f(x) = L_2$, then $L_1 = L_2$.

Suppose $L_1 \neq L_2$. Let $\epsilon = \frac{|L_1 - L_2|}{2} > 0$. There exist $\delta_1 > 0$ such that $0 < |x - a| < \delta_1 \implies |f(x) - L_1| < \epsilon$, and $\delta_2 > 0$ such that $0 < |x - a| < \delta_2 \implies |f(x) - L_2| < \epsilon$. Let $\delta = \min(\delta_1, \delta_2)$. Choose any x with $0 < |x - a| < \delta$. Then

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < \epsilon + \epsilon = 2\epsilon = |L_1 - L_2|,$$

a contradiction. Hence $L_1 = L_2$.

Section 6 The Sandwich Theorem

Theorem 8 (Sandwich Theorem or Squeeze Theorem) If $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Let $\epsilon > 0$. Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta_1 > 0$ such that if $0 < |x - a| < \delta_1$, then $|f(x) - L| < \epsilon$, which implies $L - \epsilon < f(x) < L + \epsilon$. Similarly, since $\lim_{x \rightarrow a} h(x) = L$, there exists $\delta_2 > 0$ such that if $0 < |x - a| < \delta_2$, then $|h(x) - L| < \epsilon$, which implies $L - \epsilon < h(x) < L + \epsilon$. Let $\delta = \min(\delta_1, \delta_2)$. For $0 < |x - a| < \delta$, both inequalities hold. Since $f(x) \leq g(x) \leq h(x)$, we have

$$L - \epsilon < f(x) \leq g(x) \leq h(x) < L + \epsilon.$$

Thus $L - \epsilon < g(x) < L + \epsilon$, so $|g(x) - L| < \epsilon$. Therefore, $\lim_{x \rightarrow a} g(x) = L$.

Example 10 Find $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$. Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$ for all $x \neq 0$, we have

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Both $-x^2$ and x^2 approach 0 as $x \rightarrow 0$. By the Sandwich Theorem, $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.

Example 11 Show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. For $0 < x < \frac{\pi}{2}$, consider the unit circle. The area of the sector is $\frac{x}{2}$, the area of the triangle with vertices $(0,0)$, $(1,0)$, and $(\cos x, \sin x)$ is $\frac{1}{2} \sin x$, and the area of the larger triangle with vertices $(0,0)$, $(1,0)$, and $(1, \tan x)$ is $\frac{1}{2} \tan x$. Thus

$$\frac{1}{2} \sin x < \frac{1}{2} x < \frac{1}{2} \tan x.$$

Dividing by $\frac{1}{2} \sin x > 0$ gives

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}.$$

Taking reciprocals (which reverses inequalities) yields

$$\cos x < \frac{\sin x}{x} < 1.$$

Since $\lim_{x \rightarrow 0^+} \cos x = 1$ and $\lim_{x \rightarrow 0^+} 1 = 1$, by the Sandwich Theorem, $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$. Because $\frac{\sin x}{x}$ is an even function, the left-hand limit is also 1. Hence the two-sided limit is 1.